

AI-ASSISTED INCIDENT MANAGEMENT PROGRAM PLAN

AI Incident Triage & Response Coordinator

1. Executive Summary

Objective

Design and implement an AI-powered Incident Triage & Response Coordinator that improves the effectiveness of incident management by reducing triage effort, accelerating root cause investigation, and improving incident response consistency.

The solution leverages AI capabilities to automate incident context gathering, ownership identification, historical incident correlation, runbook discovery, stakeholder communications, and post-incident reporting while maintaining human accountability for production decisions.

Business Problem

During Sev-1 and Sev-2 incidents, engineers spend significant time collecting information before they can begin troubleshooting.

Common challenges include:

- Identifying the correct responding teams
- Understanding affected systems and dependencies
- Reviewing recent deployments and changes
- Finding relevant runbooks
- Searching historical incidents
- Generating executive communications

These activities increase Mean Time To Resolution (MTTR) and create inconsistent incident response practices.

Why This Problem Was Selected

During discovery, several AI-assisted incident management opportunities were considered:

Opportunity	Benefit	Not Selected Because
Automated Remediation	High automation	Excessive operational risk
AI Chatbot for Incidents	Easy to build	Limited measurable value
RCA Generator	Useful but post-incident only	Does not reduce MTTR
Incident Triage & Response Coordinator	Direct MTTR reduction	Selected

Decision

The Incident Triage & Response Coordinator was selected because it accelerates the highest-friction activities in the incident lifecycle while maintaining human accountability for production decisions.

Desired Outcome

The proposed AI agent will:

- Reduce MTTR by 30%
- Reduce ownership identification time by 75%
- Improve incident response consistency
- Accelerate executive communications
- Improve knowledge reuse from historical incidents
- Allow engineers to focus on resolution rather than data gathering
- Improve operational resiliency
- Eliminate SRE administrative toil by automating real-time timeline scribing and post-mortem drafting

2. Problem Framing

Current Incident Workflow

Alert Generated → Incident Created → Triage → Ownership Identification → Context Gathering → War Room → Investigation → Mitigation → Recovery → RCA

Key Observation

The investigation itself is often not the longest part of an incident.

The largest delays occur during:

- Ownership identification
- Dependency discovery
- Context gathering
- Historical research

Opportunity

AI is most valuable when automating repetitive information retrieval tasks.

Rather than replacing engineers, the agent acts as an operational copilot that rapidly assembles the information required to make effective decisions.

3. User Personas

Primary User:

Incident Commander

Responsibilities:

- Coordinate incident response
- Escalate appropriately
- Communicate status
- Drive recovery

Needs:

- Fast situational awareness
- Impact assessment
- Ownership recommendations
- Executive communications

Secondary Users:

SRE Engineers

Needs:

- Service dependencies
- Runbooks
- Historical incidents
- Recent deployment history

Application Teams

Needs:

- Technical diagnostics
- Service ownership clarity
- Change history

Leadership

Needs:

- Business impact assessment
- Resolution status
- Executive summaries

4. Assumptions

Area	Assumption
Monitoring	Organizations use tools such as Datadog, Splunk, Dynatrace, or equivalent
Incident Platform	PagerDuty, ServiceNow, or similar platform is available
Knowledge Sources	Runbooks, CMDB, incident records, and change history are accessible
AI Platform	Microsoft Copilot, Azure AI, Claude, or equivalent enterprise AI solution
Security	Read-only access to incident-management systems
Human Oversight	Engineers remain accountable for all production decisions

5. Scope and Boundaries

In Scope

The AI Agent will:

Incident Correlation

- Group related alerts
- Correlate symptoms
- Reduce alert noise

Impact Assessment

- Identify affected services
- Determine customer impact
- Identify business criticality

Ownership Discovery

- Recommend resolver groups
- Identify on-call teams
- Escalate based on service ownership

Runbook Discovery

- Retrieve relevant runbooks
- Recommend troubleshooting guides

Historical Incident Analysis

- Search historical incidents
- Surface similar failure patterns
- Present prior resolutions

Communication Support

- Generate incident summaries
- Draft executive updates
- Create stakeholder communications

RCA Drafting

- Generate initial RCA documentation
 - Create incident timelines
 - Suggest action items
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Out of Scope

The AI Agent will NOT:

- Restart production services
- Deploy production changes
- Approve rollbacks
- Close incidents
- Override engineering decisions

Human operators maintain operational accountability.

6. AI Agent Design

Agent Name

AI Incident Triage & Response Coordinator

Capability 1 - Incident Context Aggregation

Automatically collects:

- Alerts
- Logs
- Metrics
- Traces
- Service dependencies

Output:

Single Incident Context Summary

Capability 2 - Ownership Recommendation

Determines:

- Primary resolver team
- Secondary support teams
- Escalation contacts

Based on:

- Service ownership
- CMDB data
- On-call schedules

Capability 3 - Change Correlation

Analyzes:

- Application deployments
- Infrastructure changes
- Configuration updates

Identifies:

Changes occurring immediately before incident onset

Capability 4 - Historical Incident Discovery

Searches:

- Previous incidents
- Prior RCAs
- Problem records

Returns:

- Similar incidents
- Root causes
- Successful mitigations

Capability 5 - Executive Communications

Generates:

- Incident status reports
- Business impact updates
- Leadership communications

Capability 6 - RCA Drafting

Generates:

- Timeline of events
- Technical summary
- Contributing factors
- Action items

Capability 7 - Incident Timeline Scribing

Automatically captures:

- Key incident events
- Decisions made
- Mitigation actions
- Status updates

Produces:

- Real-time Incident Timeline

7. Solution Architecture

Monitoring Platforms

(Datadog / Splunk / Dynatrace / SysWatch)

Incident Platform

(ServiceNow / PagerDuty)

AI Incident Agent

Knowledge Layer

- CMDB
- Runbooks
- Incident Records
- RCA Repository
- Deployment History
- Change Records
- Service Dependency Maps

Action Layer Outputs

Incident Summary

Ownership Recommendations

Impact Assessment

Runbook Suggestions

Executive Updates

RCA Draft

8. Future State Incident Workflow

Alert Generated → Incident Created → AI Triage → Ownership Recommendation → Incident Brief Generated → War Room Initiated → Investigation → Resolution → AI Draft RCA → Engineering Review → Incident Closure

9. Sample AI Output

Example Incident Summary

INCIDENT ID: INC-56789

Severity: SEV-1

Affected Service: Payment API

Customer Impact: Approximately 40% of checkout transactions impacted
Related Services: Checkout Service, Billing Service
Recent Changes: Deployment completed 15 minutes before incident onset
Recommended Responders: 1. Payments Team, 2. Database Team
Similar Incidents: INC-3342, INC-4178
Relevant Runbooks: Payment API Latency Runbook, Database Pool Exhaustion Runbook
Confidence Score: 82%
Recommended Next Step: Validate database connection utilization and recent deployment changes.

10. Program Roadmap

Phase 1 - Discovery and Design (Weeks 1-2)

Activities

- Review incident processes
- Analyze historical incidents
- Define integrations
- Define governance

Deliverables

- Requirements
- Architecture
- Success metrics

Phase 2 - Prototype Development (Weeks 3-5)

Capabilities

- Incident summarization
- Ownership recommendations
- Runbook search

Deliverables

- MVP Agent
- Test environment

Phase 3 - Pilot (Weeks 6-8)

Scope

Sev-2 incidents only

Objectives

- Validate recommendation accuracy
- Build user trust
- Measure MTTR improvements

Exit Criteria

- Recommendation accuracy $\geq 80\%$
- Positive responder feedback

Phase 4 - Production Rollout (Weeks 9-12)

Scope

- Sev-1 incidents
- Sev-2 incidents

Deliverables

- Production deployment
- User training
- Support model

Phase 5 - Optimization (Weeks 13-16)

Capabilities

- RCA generation
 - Trend analytics
 - Executive dashboards
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11. Governance Model

Meeting	Frequency	Purpose
Delivery Review	Weekly	Program execution
Steering Committee	Bi-weekly	Decisions and escalations
Pilot Review	Weekly	Accuracy and adoption review
KPI Review	Monthly	MTTR and outcome tracking

Ownership Matrix				
Activity	TPM	SRE	Security	AI Engineering
Requirements	A	C	C	R
Design	C	C	C	A
Pilot	A	R	C	R
Production Rollout	A	R	C	R
Governance	A	C	R	C

A = Accountable
R = Responsible
C = Consulted
I = Informed

Role	Responsibility
Executive Sponsor	Funding and strategic guidance
TPM	Delivery ownership and governance
SRE Lead	Incident-management alignment
Security Lead	Compliance and access controls
AI Engineering Team	Development and maintenance
Incident Commanders	Operational adoption and feedback

12. Risks and Controls

Risk	Impact	Mitigation
Incorrect ownership recommendations	Delayed response	Human validation required
Hallucinated conclusions	Incorrect troubleshooting	Evidence-based outputs only
Low engineer trust	Reduced adoption	Confidence scoring and transparency
Data privacy concerns	Compliance risk	RBAC and audit logging
Incomplete knowledge base	Poor recommendations	Content curation process
AI Model Drift	Decreasing recommendation quality over time	Quarterly model review and retraining

13. KPIs and Success Metrics

Incident Management KPIs		
Metric	Current	Target
MTTR	120 Minutes	84 Minutes
Ownership Identification	20 Minutes	< 5 Minutes
Context Gathering	30 Minutes	< 3 Minutes
Executive Updates	15 Minutes	< 2 Minutes
RCA Creation	4 Hours	< 30 Minutes
Escalation Accuracy	70%	>90%
Runbook Utilization	40%	>80%

- Adoption Metrics**
- AI Usage Rate
 - Recommendation Acceptance Rate
 - Incident Commander Satisfaction
 - SRE Satisfaction

Quality Metrics

- Recommendation Accuracy
 - False Positive Rate
 - Confidence Score Accuracy
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14. Program Success Criteria

The program will be considered successful when:

Operational Success

- Reduce Mean Time To Resolution (MTTR) by 30%
- Faster incident triage
- Improved responder productivity

Business Success

- Reduced customer-impact duration
- Improved availability
- Faster stakeholder communications

Adoption Success

- AI utilized in the majority of Sev-1 and Sev-2 incidents
- Recommendation accuracy exceeds 80%
- Positive feedback from SRE teams

Human-in-the-Loop Controls

To ensure safe AI adoption:

- AI provides recommendations only
- Human responders approve all actions
- Confidence scores accompany recommendations
- Evidence sources are displayed
- All AI recommendations are auditable

Business Outcome

- The solution should demonstrate measurable operational efficiency improvements without introducing additional production risk.
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15. Non-Negotiables

1. AI supports decision-making but never replaces human accountability.
2. All production actions require human approval.
3. All AI recommendations must provide supporting evidence.
4. AI outputs must include confidence scoring.
5. Security and compliance controls apply to all AI integrations.
6. Engineers remain accountable for incident resolution.
7. Success is measured by operational outcomes rather than AI utilization alone.

16. Final Principle

- AI accelerates incident response, but accountability for production reliability always remains with engineering and SRE teams.